(a) Zero (b) 1/6
(c) 1/2 (d) 5/6

Ans. (a) : As there are 6 letters and envelopes, so if exactly 5 are into correctly addressed envelopes, then the remaining 1 will automatically be placed in the correctly addressed envelope. Thus, the probability that exactly 5 go into the correctly addressed envelope is zero.

141. 

There are two identical red, two identical black and two identical white balls. In how many different ways can the balls be placed in the cells (each cell to contain one ball) shown above such that balls of the same colour do not occupy any two consecutive cells?

(a) 15 (b) 18
(c) 24 (d) 30

Ans. (c) : Let us start with Red colour where, R = Red, B = Black, W = White. There are eight such arrangements, if we start with Red ball similarly, there are 8 arrangements, if we start with Black or white ball.

Hence No. of arrangements = 8 + 8 + 8 = 24.

142. Which one of the following is the correct sequence in respect of the Roman numerals C, D, L and M?

(a) C > D > L > M (b) M > L > D > C
(c) M > D > C > L (d) 1 > C > D > M

Ans. (c) : Roman numerals are those used by the ancient Romans. These numbers are letters that were converted to numbers. Some Roman numerals are as follows -

L = 50, C = 100, D = 500 , M = 1000 So the correct sequence of the given Roman numerals C , D , L , M is - M > D > C > L.

143. In the series AABABCABCDABCDE .....which letter occupies the 100th position?

(a) H (b) I
(c) J (d) K

Ans. (b) : 1 12 123 1234
A, AB, ABC, ABCD----

100th word

1+2+3+4+5+6+7+8+9+10+11+12+13+[9]

It means 100th word = 9th word of alphabet= I.

144. What is the number of terms in the series 117, 120, 123, 126, ..., 333?

(a) 72 (b) 73
(c) 76 (d) 79

Ans. (b) : 117, 120, 123, 126, ..., 333

Given series is an A.P series with first term, a = 117, last term l =333 and common difference, d = 3

last term, l = a + (n – 1)d

where, n = number of terms.

117 + (n – 1)3 = 333

(n – 1)3 = 216

n = 73

145. In how many different ways can four books A, B, C and D be arranged one above another in a vertical order such that the books A and B are never in continuous position?

(a) 9 (b) 12
(c) 14 (d) 18

Ans. (b) : Books A , B , C and D can be arranged in vertical order according to the method given in the question in the following way -

ACDB BCDA

ADCB BDCA

ACBD BCAD

ADBC BDAC

CADB DACB

CBDA DBCA

So in total 12 ways can be arranged. Other methods

(i) There are 4 books, and we can arrange these 4 books in 4! ways.

(ii) If we consider book A & B always together then we can arrange the books is 3! × 2 ways.

(iii) If we house to find out the ways of arrangement in which books A & B never be together then.

Arrangement of all 4 book-arrangement of if A & B always be together.

= 4! – 3! × 2

= 4 × 3 × 2 × 1 – 3×2×2

= 24–12 = 12 ways.

146. Carpenter A can make a chair in 6 hours, carpenter B in 7 hours and carpenter C in 8 hours. If each carpenter works for 8 hours per day, how many chairs will be made in 21 days?

(a) 61 (b) 67
(c) 73 (d) 79

Ans. (c) : Chair made by carpenter A in 1 hour = 8/6
Chair made by carpenter B in 1 hour 8/7

Chair made by carpenter C in 1 hour = 8/8 = 1

So total chairs made by carpenters A, B, C in 21 days

= 21x 8/6 + 21x 8 /7 +21x 1 = 73

147. A person purchases 100 pens at a discount of 10%. The net amount of money spent by the person to purchase the pens is Rs. 600. The selling expenses incurred by the person are 15% on the net cost price. What should be the selling prices for 100 pens in order to earn a profit of 25%?

(a) Rs. 802.50 (b) Rs. 811.25
(c) Rs. 862.50 (d) Rs. 875

Ans. (c) : Buys 100 pens at 10% discount,

Net expenditure on purchase of pens = 600

Selling expenses = 15% of cost price = 15% of 600 = Rs 90.

:: . Total expenditure = Rs.690.

Profit = 25%

:: . 25% of 690 = 172.50

: . Selling price = 690 + 172.50 = Rs 862.50.

148. A school teacher has to select the maximum possible number of different groups of 3 students out of a total of 6 students. In how many groups any particular student will be included?

(a) 6 (b) 8
(c) 10 (d) 12

Ans. (c) : Suppose any particular student is always selected. Now, remaining 2 students are to be selected out of the remaining 5 students.

It can be done in 5C2 ways.

= 5!/ 2! x 3! = 10

149. In an examination, 70% of the students passed in the Paper I, and 60% of the students passed in the Paper II. 15% of the students failed in both the papers while 270 students passed in both the papers. What is the total number of students?

(a) 600 (b) 580
(c) 560 (d) 540

Ans. (a) : Percentage of total students who failed in the examination = 15 + 15 + 25 = 55 %

In addition, percentage of students who passed the total = 100 - 55 = 45

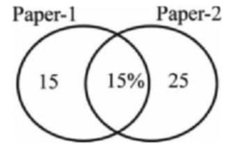
Total number of students assuming X

X x 45/100 = 270

X = 270 x 100/ 45 = 60

Total number of students on this basis = 60 x 10 = 600

Failed percentage



So total failed percentage of student

= 15 + 15 + 25 = 55%

Hence student percentage who passed in both paper = 100–55 = 45%

45% → 270 then total student

$= \frac{270}{45} \times 100 = 600$

150. March 1, 2008 was Saturday. Which day was it on March 1, 2002?

(a) Thursday (b) Friday
(c) Saturday (d) Sunday

Ans. (b) : In a year, number of weeks

= 52 extra day = 1

From 2002 to 2008, there are 6 years.

So number of extra days = 6 (1) = 6

While 2004 and 2008 are leap years,

having one more extra day apart from the normal extra day.

Thus, number of extra days = 6 + 1 + 1 = 8

Out of these 8 extra days, 7 days form a week and so 1 day remains.

Hence, March 1, 2002 is 1 day less then March 1, 2008 i.e., it is Friday.